Topic to disscuss

→ Lagrange's Inverse Interpolation

→ Numerical Problem

→ Homework Problem with solution.

Lagrange's Inverse Interpolation

x	y
x_0	y_0
x_1	y_1
x_2	y_2
x_3	y_3

$X \rightarrow ? - I$

$X \leftarrow \dot{y} = II$

Lagrange Inverse Interpolation is used when the value of the function y is known and the corresponding value of x is required.

Instead of expressing y as a function of x, the roles of x and y are interchanged, and Lagrange's interpolation formula is applied to find x in terms of y. This method is especially useful when the equation $y=f(x)$ cannot be easily solved for x directly.

Lagrange's Inverse Interpolation formula

$$x = \frac{(y-y_1)(y-y_2)(y-y_3)}{(y_0-y_1)(y_0-y_2)(y_0-y_3)}x_0 + \frac{(y-y_0)(y-y_2)(y-y_3)}{(y_1-y_0)(y_1-y_2)(y_1-y_3)}x_1$$

$$+ \frac{(y-y_0)(y-y_1)(y-y_3)}{(y_2-y_0)(y_2-y_1)(y_2-y_3)}x_2 + \frac{(y-y_0)(y-y_1)(y-y_2)}{(y_3-y_0)(y_3-y_1)(y_3-y_2)}x_3$$

Numerical Problem : 1

Q: Use Lagrange's Inverse Interpolation formula to find the value of x, when $y = 0.143$ from the following data.

x	1	2	4	5	8
$f(x)$	1.000	0.500	0.250	0.200	0.125

Solution, Given, $y = 0.143$

Applying Lagrange's inverse formula,

	x_0	x_1	x_2	x_3	x_4
x	1	2	4	5	8
$f(x)$	1.000	0.500	0.250	0.200	0.125
	y_0	y_1	y_2	y_3	y_4

$y = 0.143$

$$x = \frac{(y-y_1)(y-y_2)(y-y_3)(y-y_4)}{(y_0-y_1)(y_0-y_2)(y_0-y_3)(y_0-y_4)}x_0 + \frac{(y-y_0)(y-y_2)(y-y_3)(y-y_4)}{(y_1-y_0)(y_1-y_2)(y_1-y_3)(y_1-y_4)}x_1 + \frac{(y-y_0)(y-y_1)(y-y_3)(y-y_4)}{(y_2-y_0)(y_2-y_1)(y_2-y_3)(y_2-y_4)}x_2$$

$$+ \frac{(y-y_0)(y-y_1)(y-y_2)(y-y_4)}{(y_3-y_0)(y_3-y_1)(y_3-y_2)(y_3-y_4)}x_3 + \frac{(y-y_0)(y-y_1)(y-y_2)(y-y_3)}{(y_4-y_0)(y_4-y_1)(y_4-y_2)(y_4-y_3)}x_4$$

$$x = \frac{(0.143-0.500)(0.143-0.250)(0.143-0.200)(0.143-0.125)}{(1.000-0.500)(1.000-0.250)(1.000-0.200)(1.000-0.125)}\times 1 + \frac{(0.143-1.000)(0.143-0.250)(0.143-0.200)(0.143-0.125)}{(0.500-1.000)(0.500-0.250)(0.500-0.200)(0.500-0.125)}\times 2$$

$$+ \frac{(0.143-1.000)(0.143-0.500)(0.143-0.200)(0.143-0.125)}{(0.250-1.000)(0.250-0.500)(0.250-0.200)(0.250-0.125)}\times 4 + \frac{(0.143-1.000)(0.143-0.500)(0.143-0.250)(0.143-0.125)}{(0.200-1.000)(0.200-0.500)(0.200-0.250)(0.200-0.125)}\times 5$$

$$+ \frac{(0.143-1.000)(0.143-0.500)(0.143-0.250)(0.143-0.200)}{(0.125-1.000)(0.125-0.500)(0.125-0.250)(0.125-0.200)}\times 8$$

$$x = -0.000149304 + 0.013380718 - 1.071457874 + 3.2736543 + 4.85274024$$

$x = 7.068168$ Ans.

Homework Problem :-

Q: Use Lagrange's inverse interpolation to find x when $y = 12$ from

x	1	2	3
y	8	15	24

Solution : Given , $y = 12$

Applying Lagrange's Inverse Interpolation,

$$x = \frac{(y-y_1)(y-y_2)}{(y_0-y_1)(y_0-y_2)} x_0 + \frac{(y-y_0)(y-y_2)}{(y_1-y_0)(y_1-y_2)} x_1 + \frac{(y-y_0)(y-y_1)}{(y_2-y_0)(y_2-y_1)} x_2$$

$$= \frac{(12-15)(12-24)}{(8-15)(8-24)} \times 1 + \frac{(12-8)(12-24)}{(15-8)(15-24)} \times 2 + \frac{(12-8)(12-15)}{(24-8)(24-15)} \times 3$$

$$= \frac{9}{28} + \frac{32}{21} - \frac{1}{4}$$

$$= \frac{27+128-21}{84} = \frac{134}{84} = \frac{67}{42}$$

$$x = 1.595238$$

Ans.

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